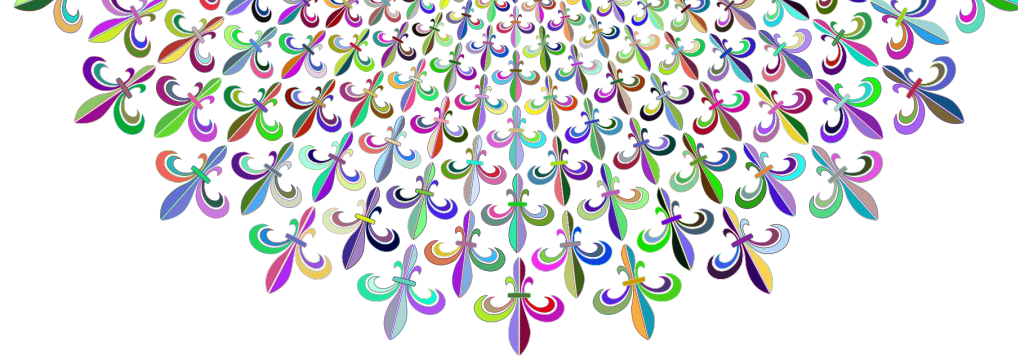




# Image Restoration

**Lucien Heitz, University of Zurich**  
Bruno N. Sotic, University of Amsterdam  
Qi Bi, Utrecht University  
Jaap Kamp, University of Amsterdam



MediaEval Workshop  
15-16 June 2026,  
Amsterdam, Netherlands  
and Online

# Background and Motivation

- Reusing historical images is important for archival purposes & augmenting existing collections
- Historic photos are low-resolution, monochromatic and physically degraded
- **RQ:** Do restored photos improve perceived image-article fit compared to the originals?



# Data

- 100 scans of newspapers
- 19th and 20th century
- BW and degraded



ID 8526, “Fifty thousand demonstrators in Lisbon”



# Approaches

## Flux.2 Klein

4B base model

Camera imperfections

## Qwen 2509

7B base model

Modern photo look

## FireRed 1.1

Model & LoRA

Modern photo look



All models receive the same degraded input and a short restoration prompt.

# Evaluation and Metrics



Offline metrics:  
NIQE, BRISQUE, NRQM, MUSIQ, ...

Assess perceptual quality and prompt alignment



Online studies:  
experts (n=18) & non-experts (n=30)

5-point Likert scale to rate headline-image fit  
Evaluation criterion is average of article fit

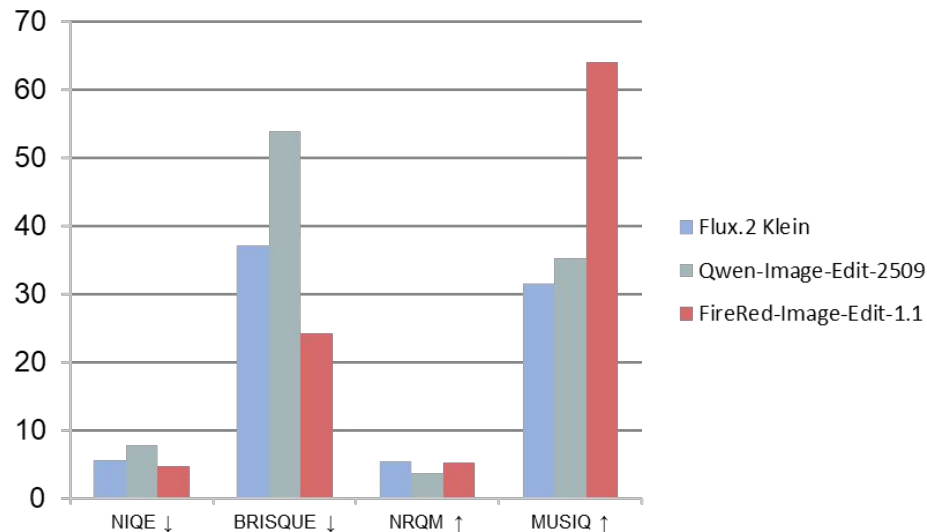


# Offline Evaluation - Perceptual Quality

FireRed achieves the best perceptual quality

Flux scores well on NIQE but lower on MUSIQ

Qwen lags behind on perceptual metrics

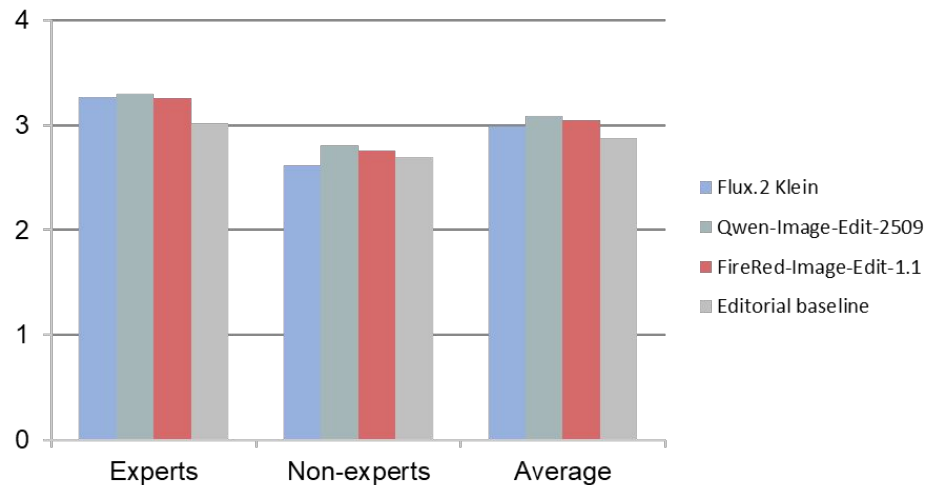


# Online Evaluation - User Studies

Experts rate images higher than non-experts

Qwen achieves top ratings in both groups

All restorations except Flux beat the editorial baseline



# Qualitative Comparison



Original



Flux.2 Klein



Qwen 2509



FireRed 1.1



Restored photos are more colorful and detailed than the original scans.



# Discussion & Insights



## **Offline metrics favour FireRed & Flux**

User studies prefer Qwen for image fit  
Perceptual metrics alone do not capture human relevance



## **Generative restoration can misrepresent reality**

Models may swap ethnicity, insert incorrect text  
Requires human oversight, disclosure & improved offline metrics



# Conclusion & Outlook



## **Restoration improves perceived image-article fit**

Experts & non-experts prefer restored photos over originals



## **Offline metrics do not reflect human judgement**

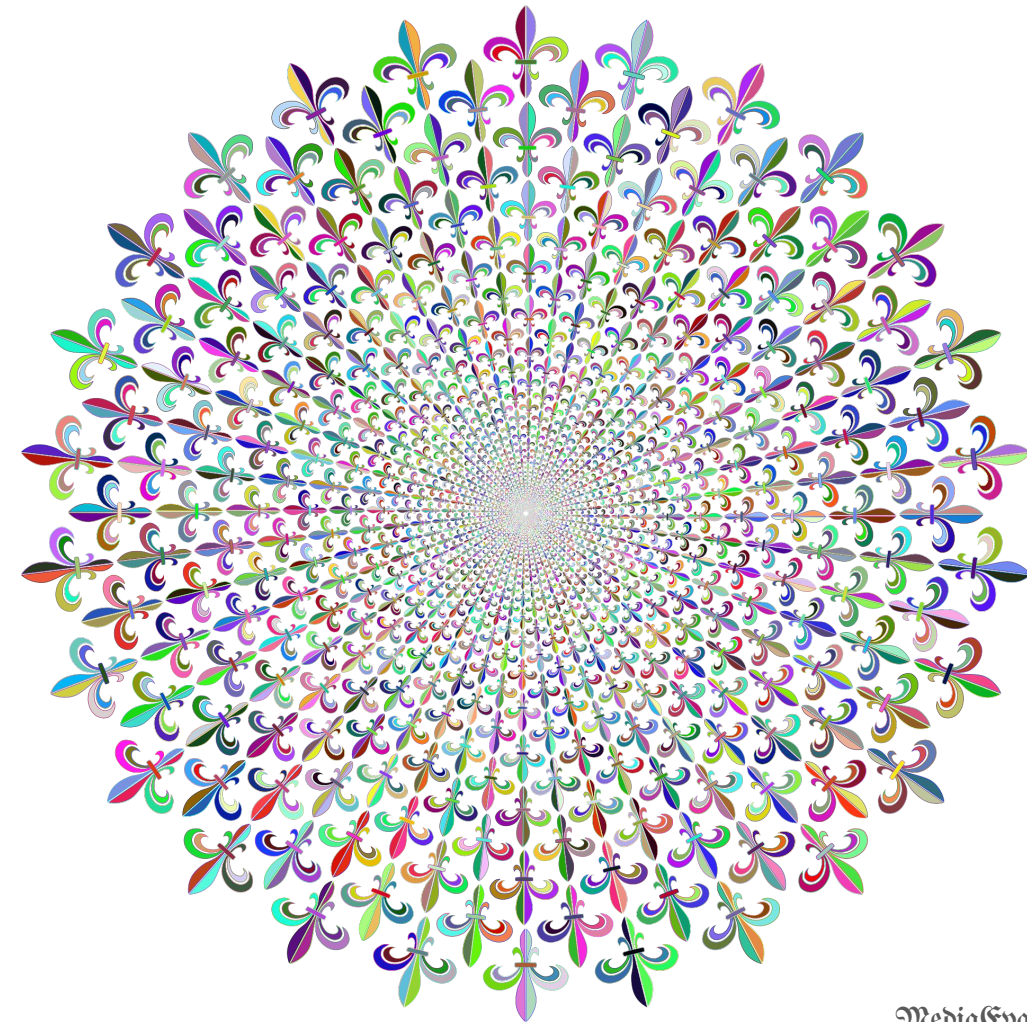
Need better no-reference metrics that correlate with relevance



## **Ethics: transparency, consent & cultural sensitivity**

Develop workflows with human oversight; expand to diverse datasets





# Questions?